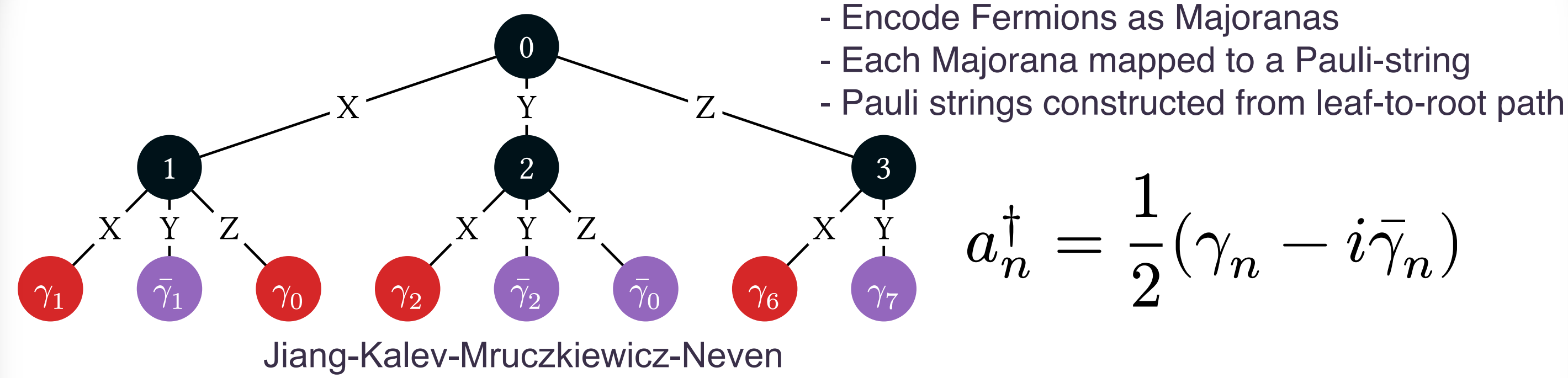


Motivation

- Digital quantum simulation of fermions requires encoding fermionic operators as qubit operators
- Choice of encoding can have a very big impact on circuits and results

Ternary Tree Encodings

- Includes Jordan-Wigner, Parity, Bravyi-Kitaev
- Node for each physical qubit
- Encode Fermions as Majoranas
- Each Majorana mapped to a Pauli-string
- Pauli strings constructed from leaf-to-root path

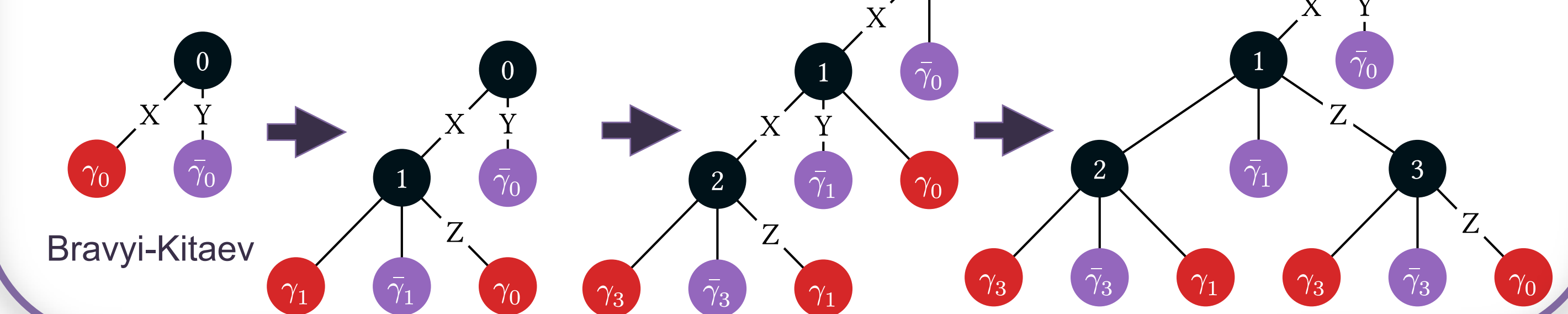


Vacuum preservation $|\rangle_f \rightarrow |0\rangle_q^{\otimes N}$

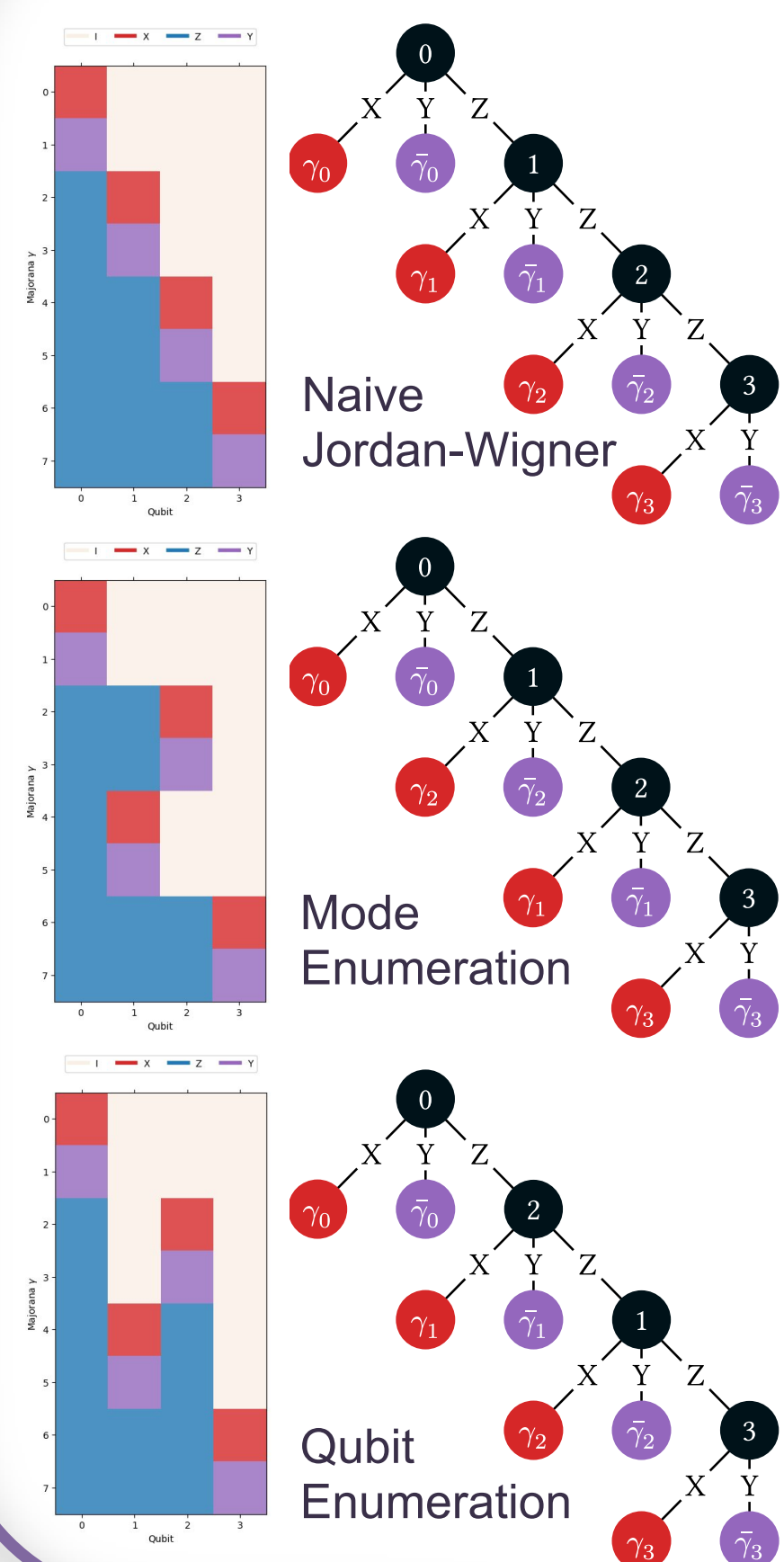
- Start with root node
- Every node gets a pair of indices, even on X, odd on Y
- New child nodes move existing leaf to Z-edge

$$a_n^\dagger = \frac{1}{2}(\gamma_n - i\bar{\gamma}_n)$$

$$a_n = \frac{1}{2}(\gamma_n + i\bar{\gamma}_n)$$



Enumeration



Degrees of Freedom

- Qubit labels
- Majorana labels

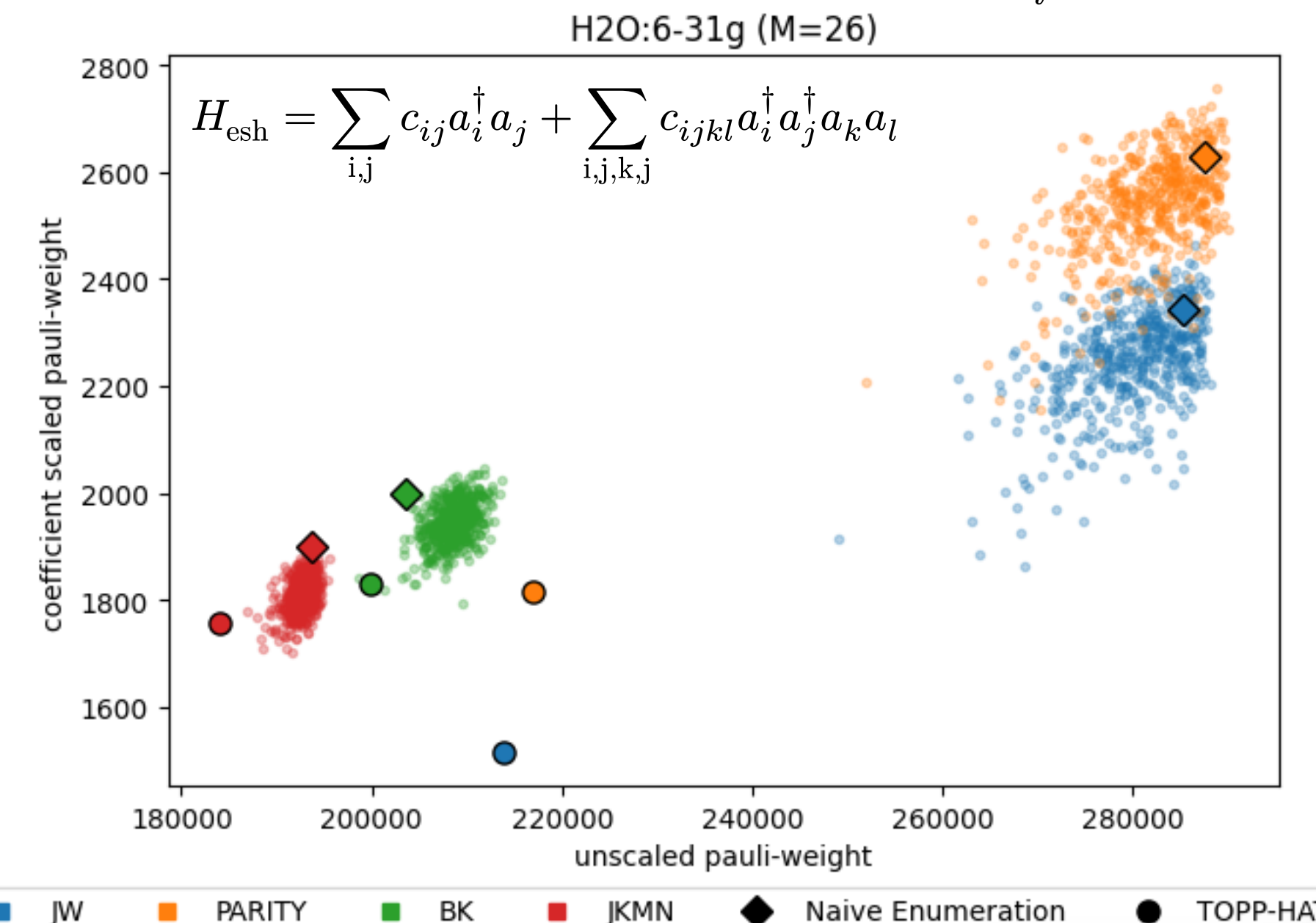
$$H_q = \sum_i c_i h_i^{\otimes n}$$

$$W_P(H_q) = \sum_i W_P(h_i)$$

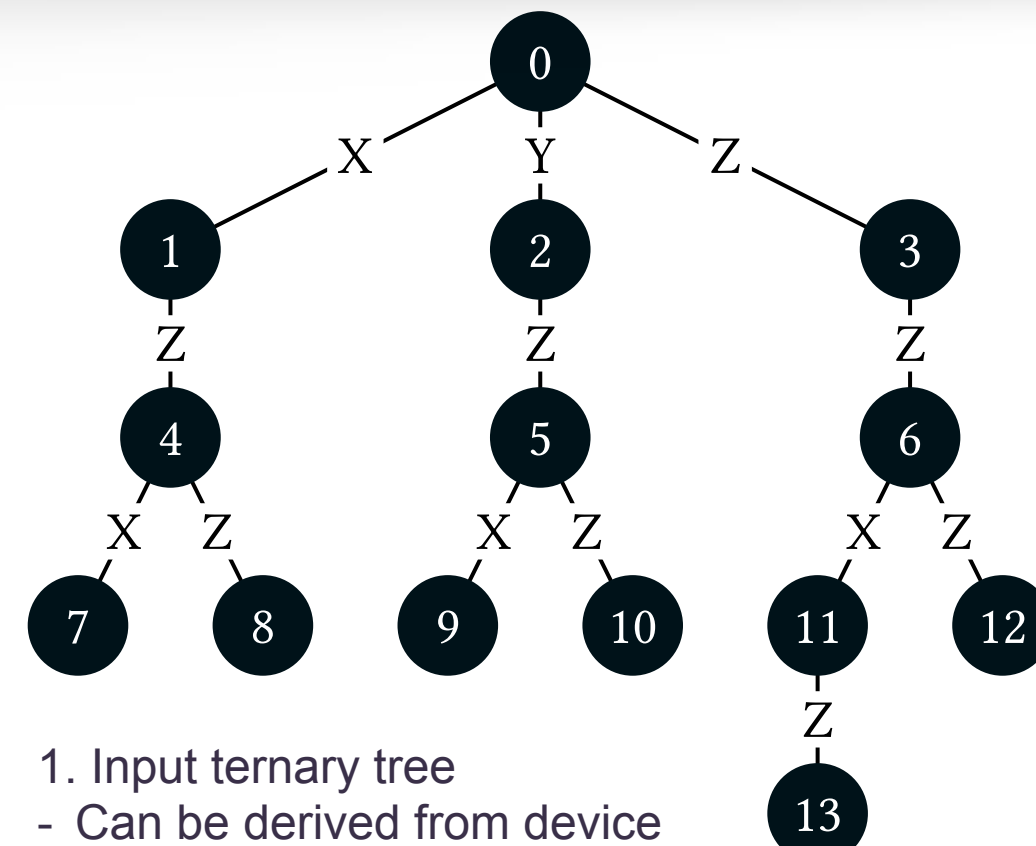
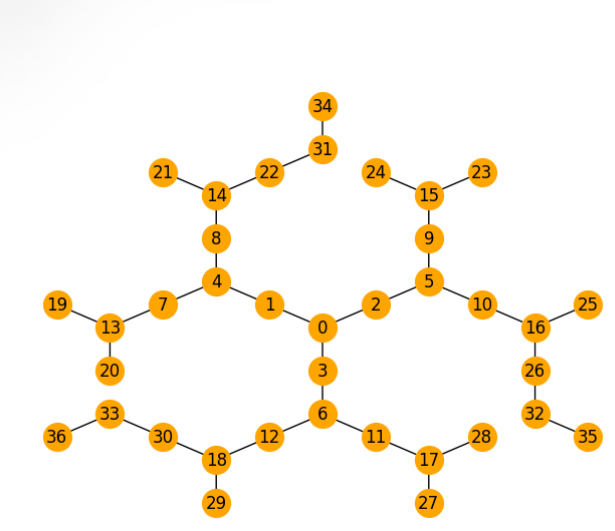
Cost Functions

- Pauli-weight
- coefficient Pauli-weight

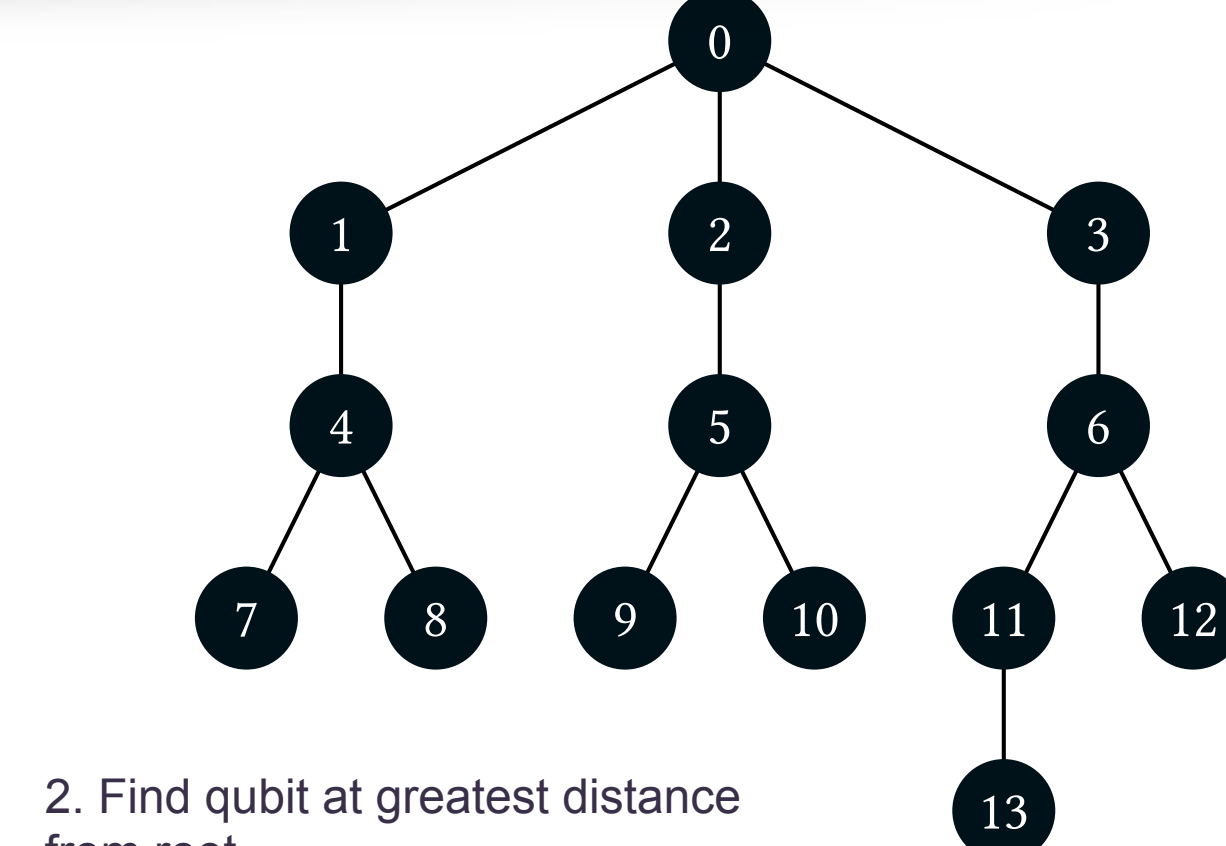
$$W_{CP}(H_q) = \sum_i |c_i| W_P(h_i)$$



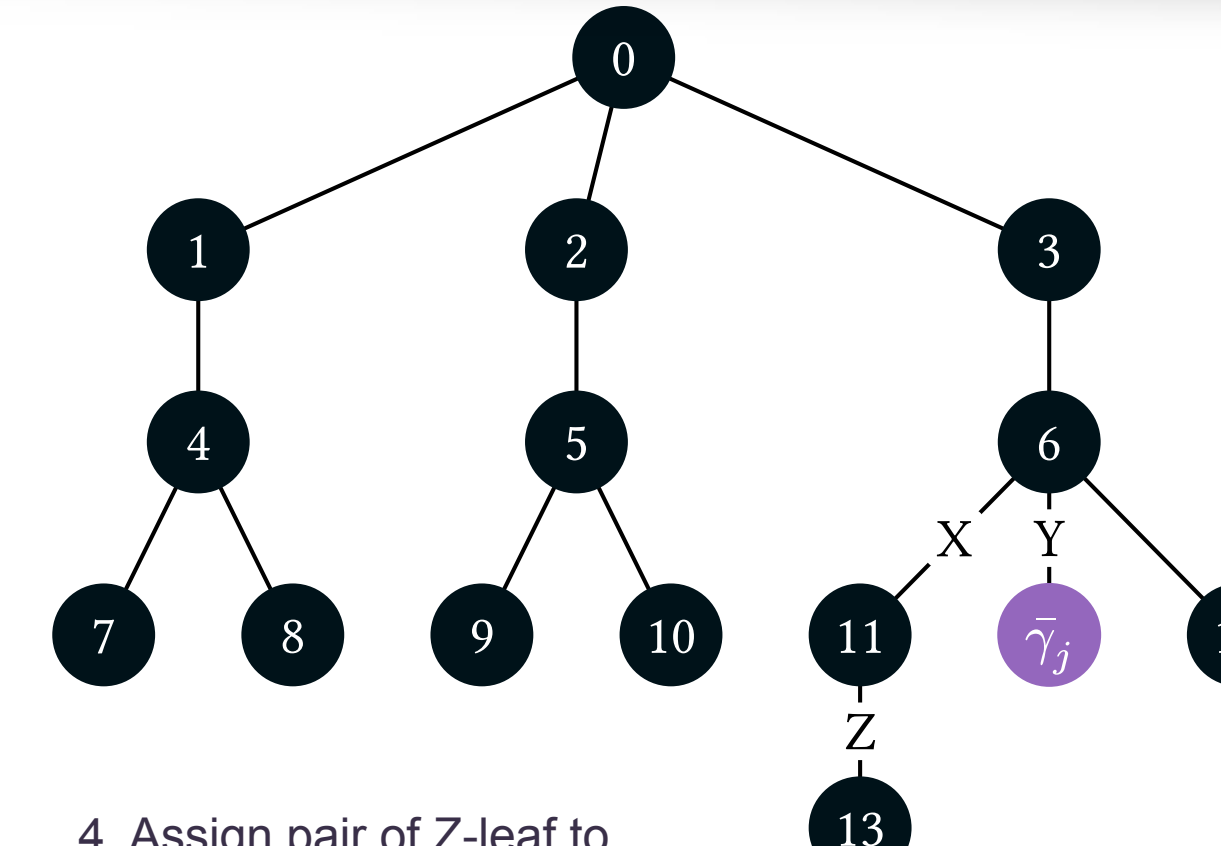
TOPP-HATT



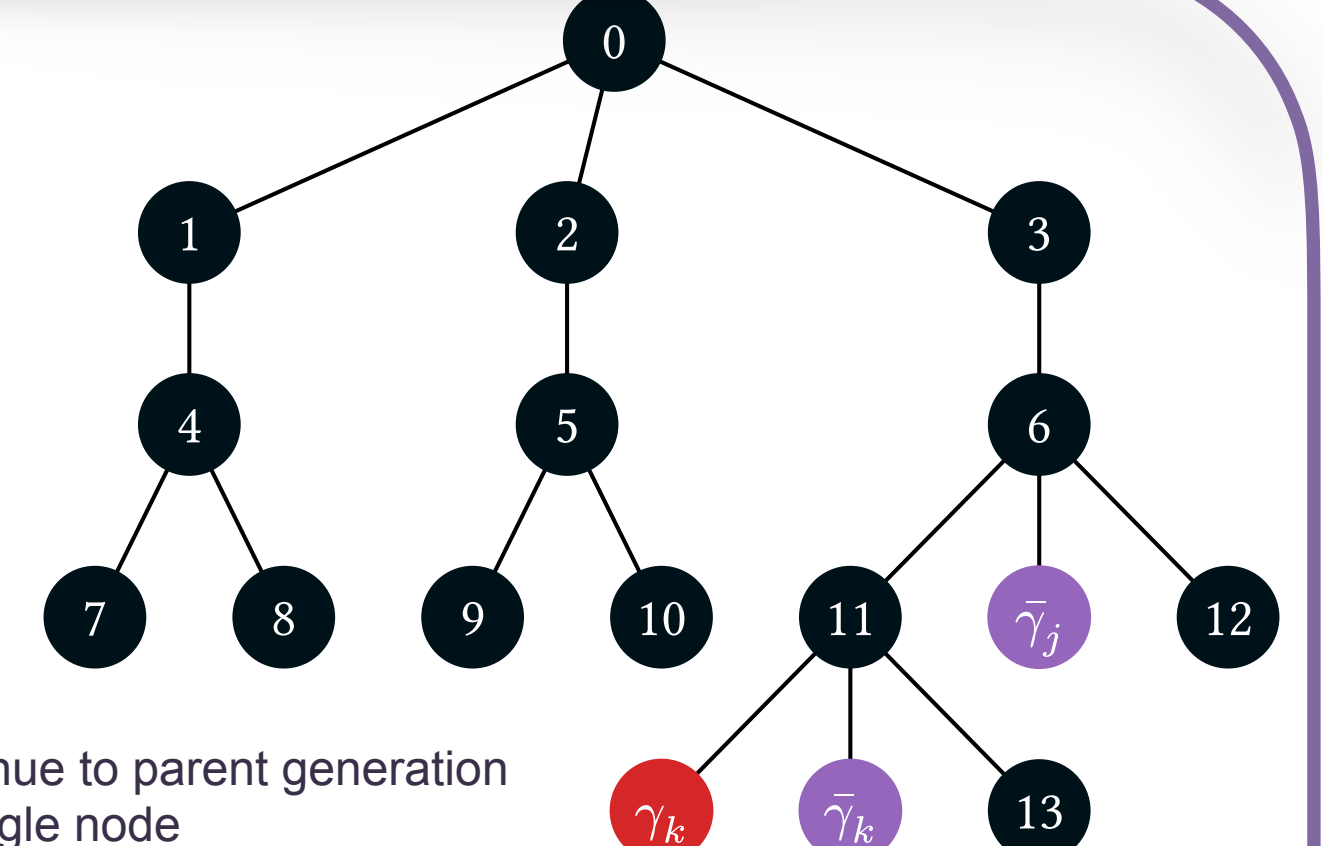
1. Input ternary tree
- Can be derived from device connectivity using Bonsai
- Can use JW, BK, Huffman, HATT etc
- Output tree remains the same shape!



2. Find qubit at greatest distance from root.
- If more than one, take X-most
3. Find Majorana indices with minimum Pauli-weight on that qubit.



4. Assign pair of Z-leaf to maintain vacuum preservation.



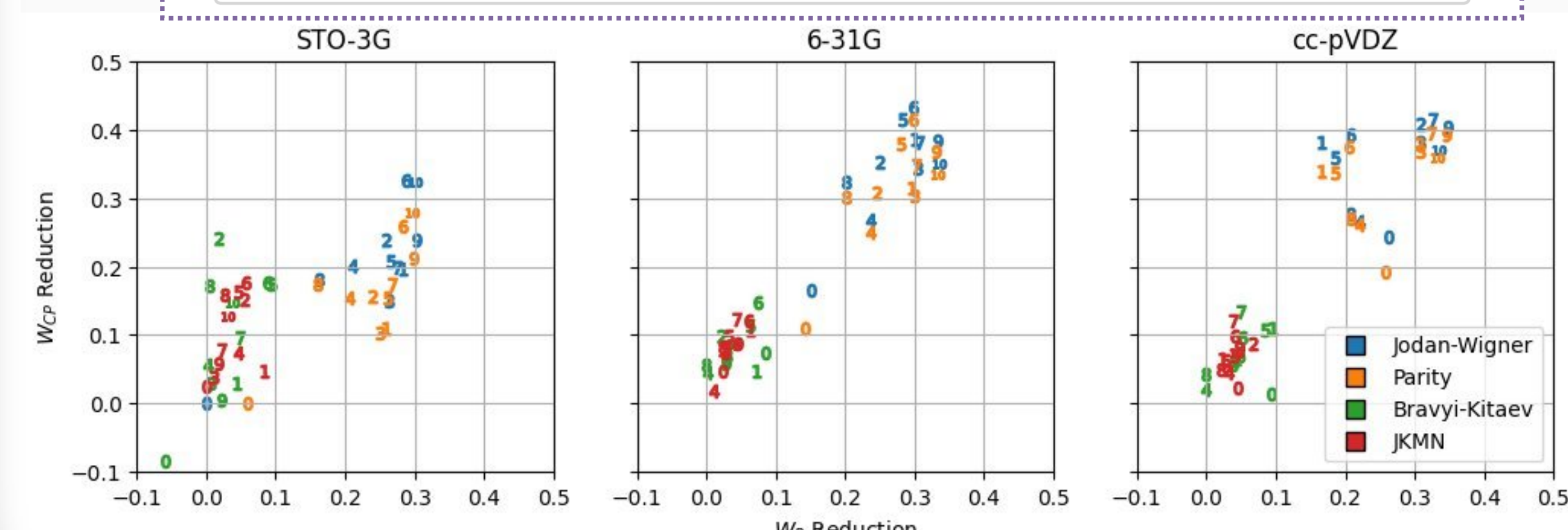
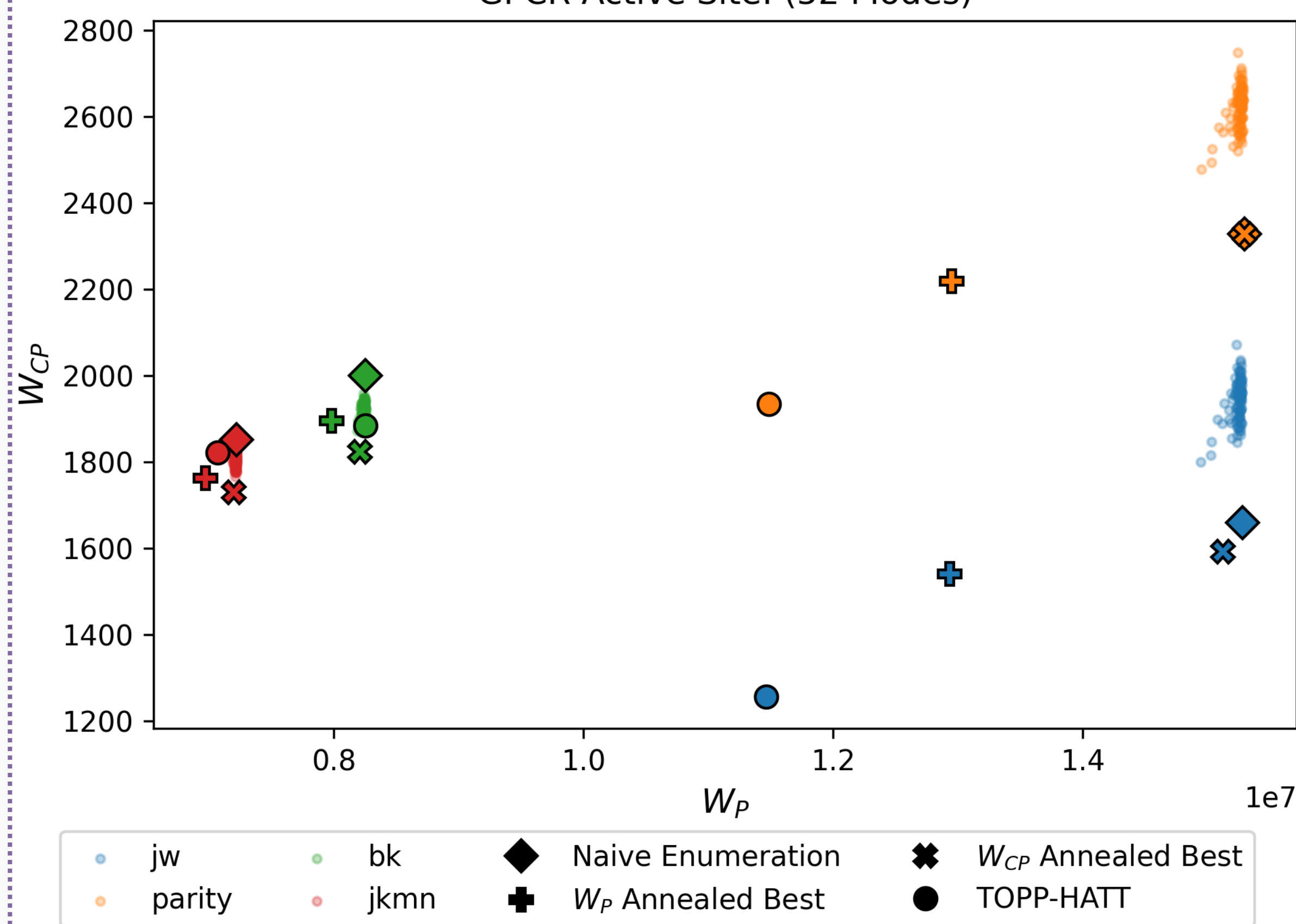
- ... continue to parent generation
- Fill single node
- Assign Z-leaf pair (if it exists)
- Repeat until generation complete
- Repeat for all generations

(the actual algorithm is ~800 lines of performance optimised rust, ask me about it!)

Weight Reduction

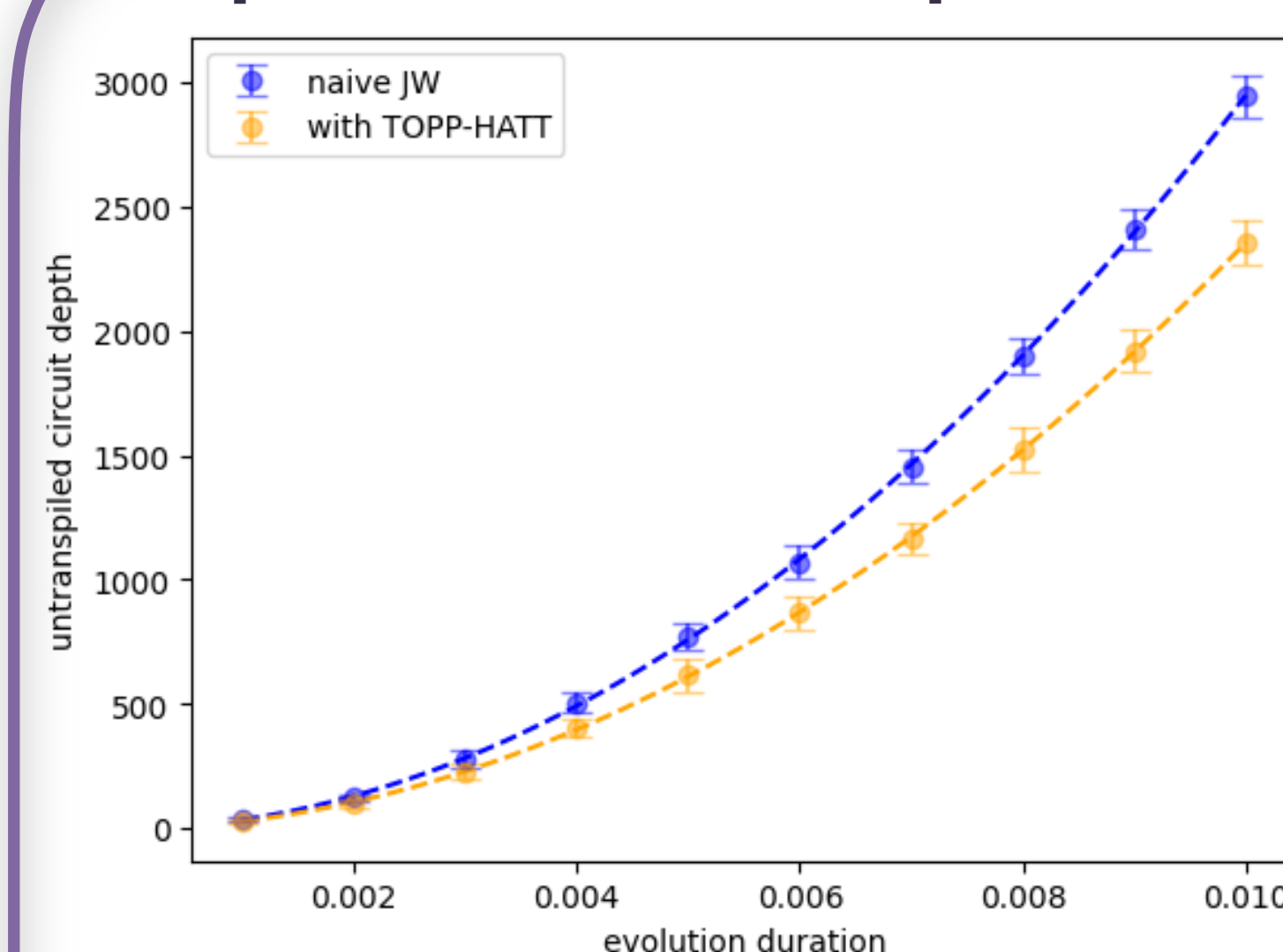
- Total weight of encoded hamiltonians can be significantly reduced.
- Best results for large molecules, using Jordan-Wigner or Parity encoding.
- Non-linear trees have less potential to optimise.

GPCR Active Site: (52 Modes)



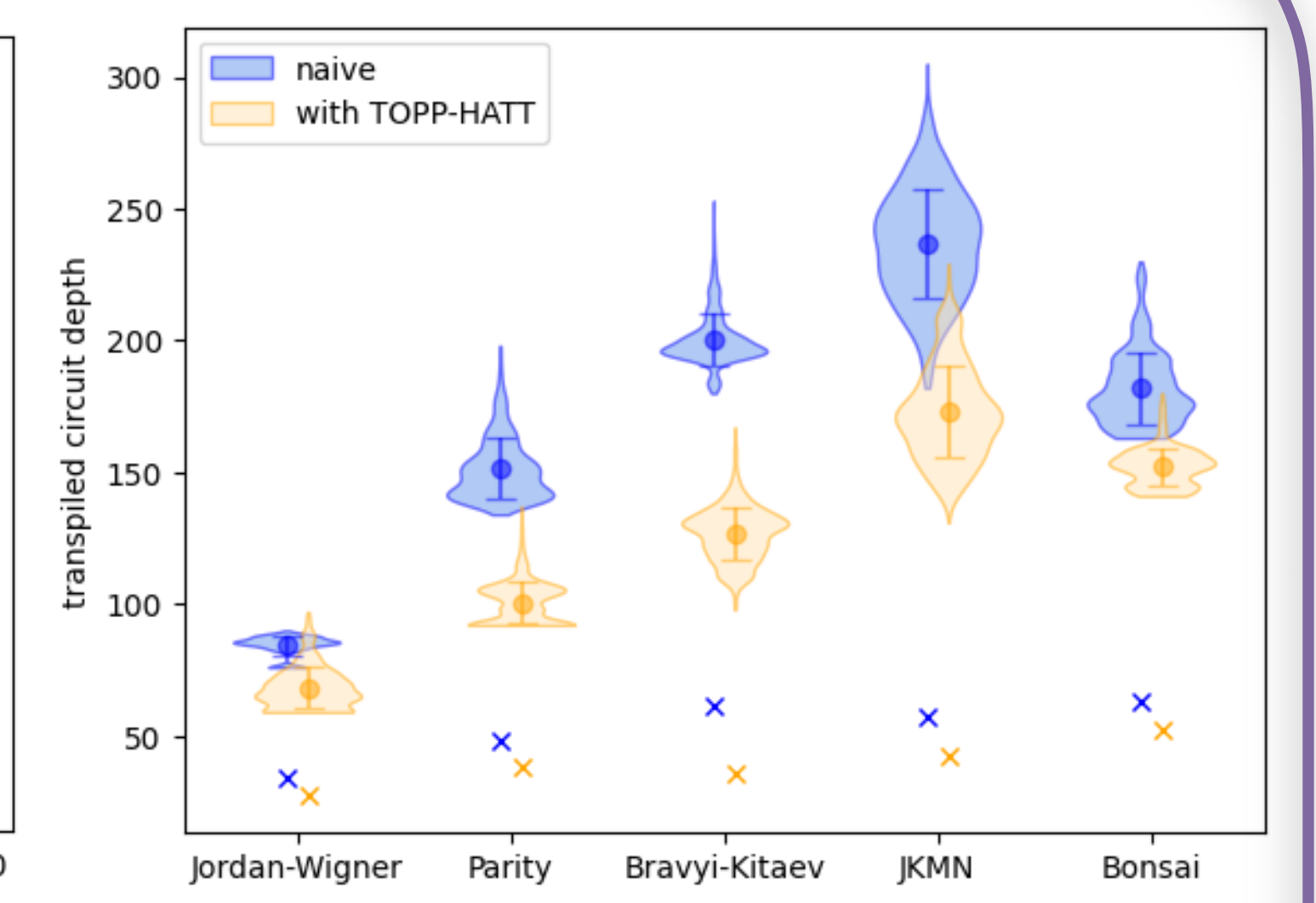
0: H2, 1: LiH, 2: H2O, 3: BH3, 4: CH4, 5: N2, 6: HCN, 7: C2H2, 8: CH3F, 9: Ethene, 10: Ozone

qDRIFT Circuit Depth Reduction



Un-transpiled qDRIFT

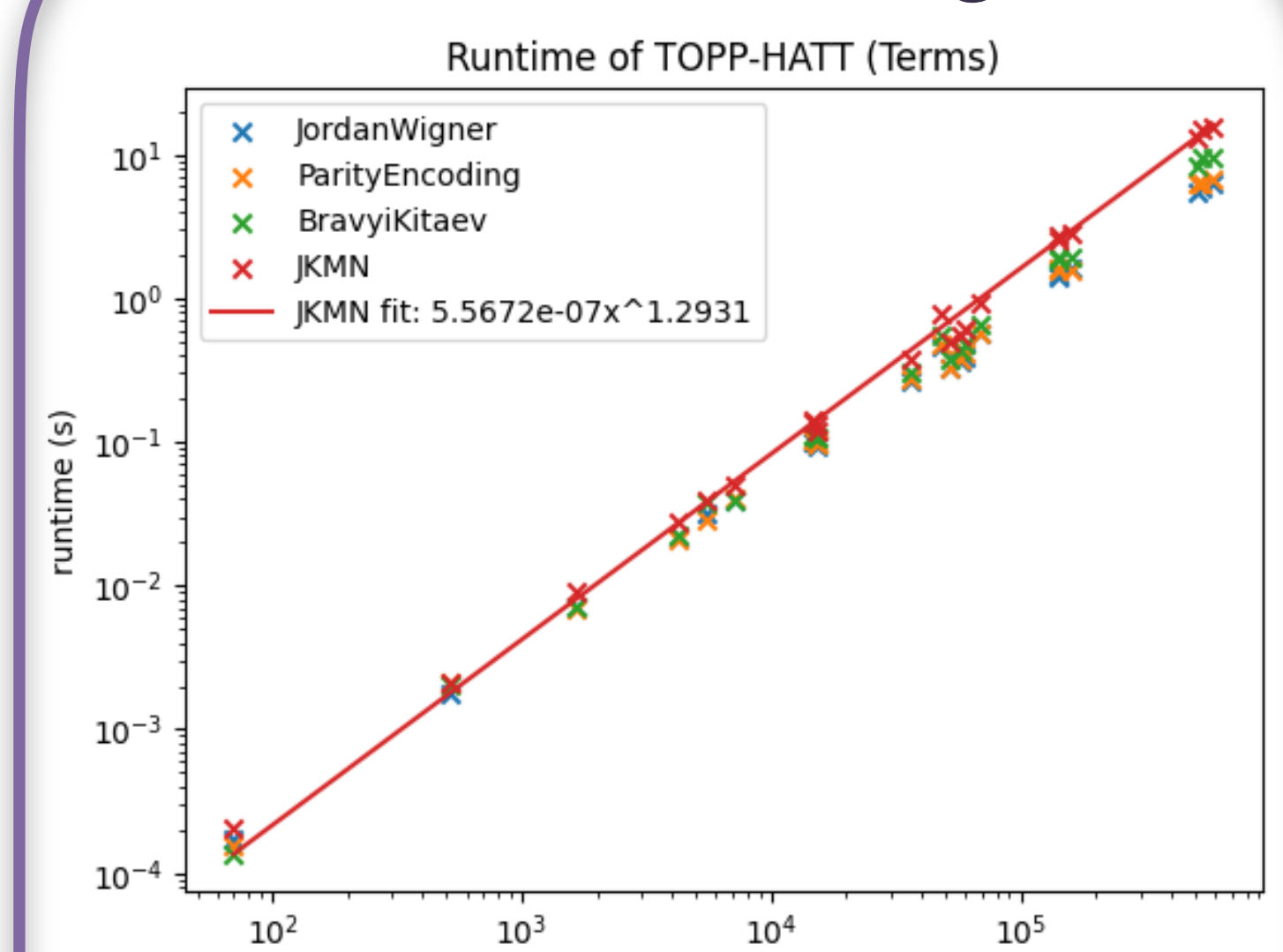
- H2O (STO-3G)
- Reduction by constant factor
- 25-30% across various encodings



Transpiled qDRIFT circuits

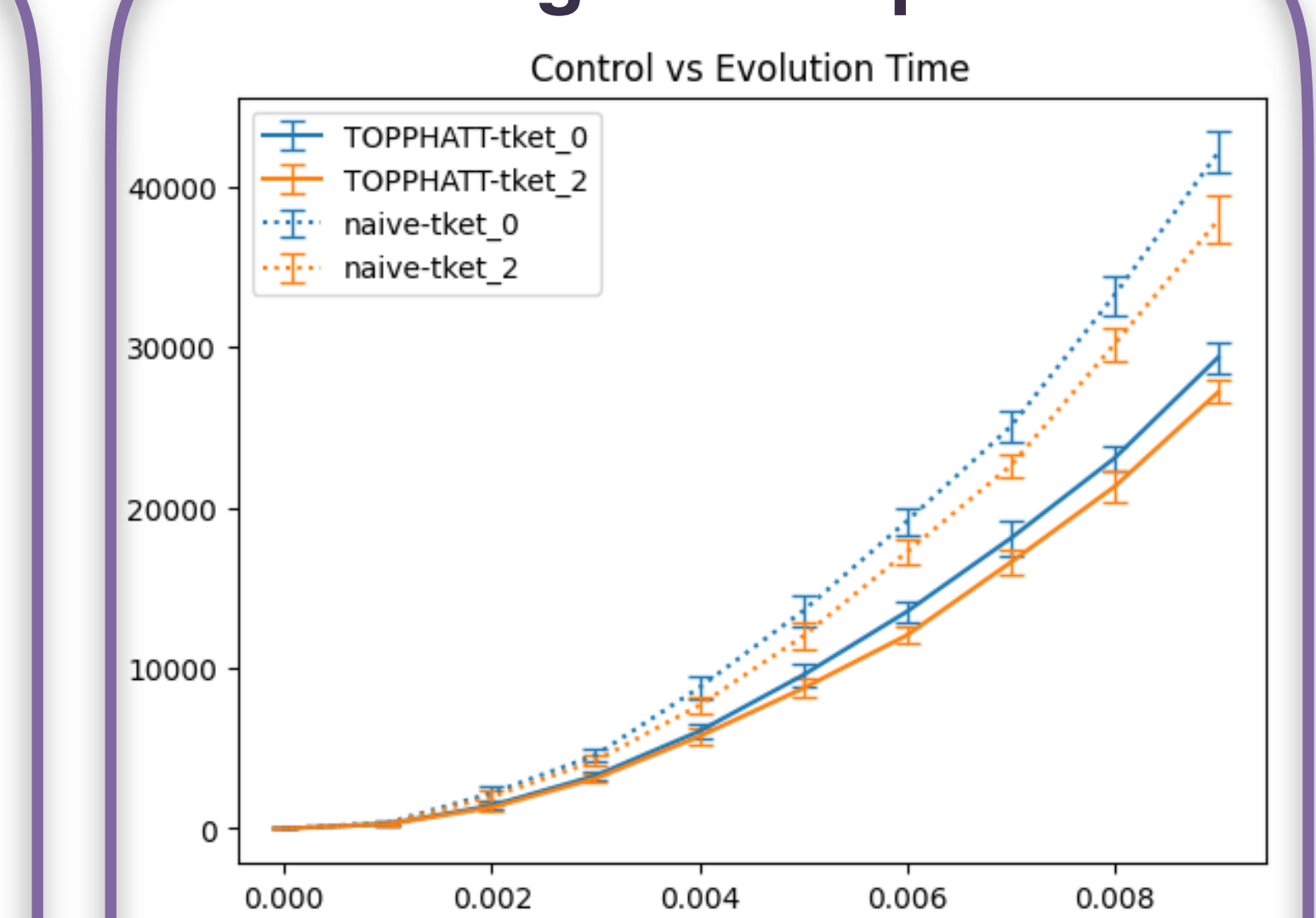
- From average depth un-transpiled
- Targets IQM Garnet (square lattice)
- Optimisation improves transpilation

Resources & Scaling



- Scales with $|H_\gamma|^{1.3}$ in worst case.
- 50 qubits in ~10 seconds
- Runtime not circuit-dependent

Encoding as compilation



Algorithms are defined in fermionic operators, let a compiler choose the best encoding!

- Encoding optimisation:
- Pairs with gate-based compilers
- Significantly improves CX count & depth

with thanks to

arXiv Preprint



Code Repository



Tutorial Notebooks

